

Radar-Only, Camera-Gated, and Emergency-Fallback AEB: Scenario-Level Trade-offs and Failure Severity in CARLA

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Abstract—Camera permission gating can suppress radar-triggered false braking but veto hazards outside a detector’s label space. We compare radar-only, hard camera gating, and camera gating with a conservative radar emergency fallback in a sensor-to-brake automatic emergency braking (AEB) pipeline in CARLA 0.9.11. Unlike a run-count analysis, the primary unit is a named condition: 105 core conditions and a 14-condition frozen adverse hold-out, each repeated five times only to check consistency. On core, radar-only, hard-gate, and fallback policies respectively obtain scenario-level precision/recall of 0.913/0.988, 1.000/0.965, and 1.000/0.988; fallback passes 101/105 conditions. This does not generalize as dominance. Hard gating passes 11/14 hold-out conditions versus 7/14 for fallback because four convincing synthetic-ghost conditions pass the fallback rules. Those fallback false brakes are severe in simulation: all 20 repeated runs stop from a median 78.4 km/h, with 3.60-s braking and 9.02 m/s² median peak deceleration. Physical-prop collisions further show why brake recall is insufficient: on an unseen bench, radar-only reduces the last pre-impact speed to 32.57 km/h, while delayed fallback reaches 54.93 km/h and hard gating 59.95 km/h. A frozen 2,461-run CUDA campaign, component ablation, perturbations, detector disabling, explicit scoring, and traceable logs support an empirical map of precision–recall and failure severity—not a general fusion advantage or safety certification.

Index Terms—automatic emergency braking, radar–camera gating, CARLA, false braking, precision–recall, fault injection, scenario-level evaluation

I. Introduction

AEB closes a perception–decision–actuation loop in which a sensor detection must become a timely and justified brake. Radar supplies range and radial velocity but can select non-hazards or spurious returns. Camera semantics can reject implausible radar targets, yet a missed or out-of-label object is not proof of free road. Reviews therefore treat sensing, target selection, risk, and control as an interdependent chain [1].

Radar–vision AEB recognition and noise filtering are established [2], [3]. A recent preprint specifically studies radar-guided camera verification [4]. This paper does not claim a new fusion primitive. It asks a narrower empirical question: can a late radar emergency override recover non-vehicle hazards rejected by a car-only camera gate without restoring radar false brakes, and which failure mechanisms prevent that goal?

Simulation enables repeatable near-collision tests [5], [6], but two practices can overstate evidence: using privileged

actor truth online, and treating deterministic repetitions as independent samples. We use only CARLA radar points and RGB frames for braking, reserve actor truth for offline labels, freeze policy and suite before hold-out, and make named conditions the primary unit. Repetitions measure process consistency. We also augment binary brake/collision counts with false-brake duration/deceleration and last-tick pre-impact speed.

The contributions are: (1) a paired comparison of radar-only, hard gate, and emergency fallback; (2) scenario-level confusion, Wilson descriptors, and paired system outcomes; (3) ablation, one-factor sensitivity, local perturbation, detector-disabled, and frozen adverse hold-out tests; and (4) a 2,461-run resumable GPU protocol with provider, hash, tick, and failure-severity traceability. The central result is both positive and negative: fallback combines the two core operating points, while high-support ghosts and late or missing physical tracks preclude a dominance claim.

II. Related Work and Positioning

Kim and Song combine radar and vision for AEB vehicle recognition [2]; Hsu *et al.* filter radar/camera noise for target selection [3]. Bae *et al.* estimate the closest in-path vehicle through LiDAR projection and camera tracking on a real vehicle [7]; Zhang *et al.* combine curved-road target selection, TTC, safety distance, and graded braking [8]. CenterFusion illustrates learned radar–camera association for 3-D detection [9], whereas our policies keep radar kinematics and camera permission explicitly separable for controlled substitution.

CARLA supports protocol-derived pipeline evaluation [10]; recent CARLA AEB work also evaluates alternative visual sensing [11]. Virtual radar fidelity depends strongly on whether the model represents objects, detections, or raw signals [12]. Our contribution is therefore a falsification-oriented late-policy comparison, not a claim that point-level CARLA radar reproduces multipath prevalence. Standards motivate scenario structure but are not reproduced as compliance tests [13], [14].

Two distinctions position the present study. First, object-level learned fusion usually seeks detection accuracy, while an AEB permission gate asks whether an already collision-relevant radar track may actuate. A missed camera association therefore has asymmetric cost: it may prevent a false brake or veto a genuine hazard. Second, simulation papers often

TABLE I
Frozen sensing and detector setup.

Item	Setting
Camera	RGB, 1280×720, 70° FOV; $(x, z) = (0.43, 1.35)$ m
Radar	100 m, 30° × 6°, 2000 points/s; $(x, z) = (2.53, 0.48)$ m
Detector	YOLO26n ONNX [15], 640 input, confidence 0.25, NMS IoU 0.50
Dataset	1505/300/200 train/val/test images; 28/6/4 disjoint collection sessions
Training	one <i>car</i> class, AdamW, seed 2026; ONNX SHA-256 prefix <code>dc0a6ca1754a</code>
Runtime	CUDA required, 0.15-s inference cadence, 0.35-s confirmation hold

report nominal success without an adverse set frozen before evaluation. Here the same attributes used to justify fallback—persistence, centrality, support, and critical risk—are deliberately reproduced by a synthetic fault after policy freeze. The contribution is the resulting policy/failure comparison and audit trail, rather than a new detector, tracker, or standards procedure.

III. System and Policies

A. Information boundary and configuration

The ego Tesla Model 3 runs on Town04 in synchronous 0.05-s steps. Table I reports the frozen sensor and detector settings. Camera and radar both tick at 0.05 s; YOLO inference is scheduled by simulation timestamp at 0.15 s, and confirmation is held for 0.35 s. Online code does not query actor identity, class truth, target velocity, or ground-truth boxes.

Dataset splits contain 1872/379/264 instances, no exact cross-split image duplicates, and 21.2/16.3/18.0% empty frames. Split session identifiers are disjoint, rather than interleaving adjacent frames across splits, although all imagery remains Town04 in-domain. Validation precision, recall, mAP₅₀, and mAP_{50:95} are 0.9919, 0.9642, 0.9940, and 0.9495; these do not establish out-of-domain safety. Exact radar-to-camera transforms project a selected target into the current image. A confirmation requires its projected point inside a retained car box after confidence and NMS filtering. Sensor callbacks carry simulation-frame timestamps; inference cadence and hold expiry use simulation rather than host time. The 0.35-s hold intentionally bridges the 0.15-s inference cadence, but also defines a bounded stale-semantic window that is identical for both camera policies.

Radar points are filtered by range, road height, and a predicted constant-curvature path, clustered by position/velocity, and associated across frames. A target track must be confirmed, non-stale, and collision-relevant. For distance d , relative radar velocity $v_r < 0$, and ego speed v_e ,

$$v_c = -v_r, \quad \text{TTC} = d/v_c, \quad v_t = \max(0, v_e + v_r). \quad (1)$$

The stopping requirement and margin are

$$d_{\text{req}} = \max\left(d_0, v_e t_r + \frac{v_e^2}{2a_e} - \frac{v_t^2}{2a_t} + d_0\right), \quad m = d - d_{\text{req}}, \quad (2)$$

with $t_r = 0.20$ s, $a_e = 8$, $a_t = 6$ m/s², and $d_0 = 1$ m as simulation risk parameters. A staged PID maps risk to continuous CARLA brake.

B. Three late brake policies

Radar-only passes the radar AEB decision B_r . Hard gating requires current camera confirmation C or held confirmation H . The emergency policy is

$$B_f = B_r \wedge (C \vee H \vee E_r). \quad (3)$$

E_r requires a confirmed/non-stale track, age and hit streak ≥ 3 , at least six cluster points, confidence ≥ 0.70 , absolute path offset ≤ 0.65 m, TTC ≤ 1.10 s, and margin $m \leq -2.0$ m. TTC and margin are conjunctive. Once accepted, fallback latches until radar exits **BRAKE**. A camera box cannot create kinematic risk without a radar target. Thus all policies share radar processing, equations, PID, scenario control, and scoring; only brake permission differs.

The fixed online order is: update ego path; filter points; update and confirm tracks; select one radar target; evaluate TTC/margin; process the latest camera frame when cadence permits; apply the selected permission policy; command the staged PID; and log the following simulator state. This boundary prevents a camera box from inventing range or relative velocity and prevents actor truth from repairing a weak radar track. It also localizes policy differences: a missing cluster fails before every gate, while a confirmed but unlabelled track separates hard gate from fallback.

IV. Frozen Protocol and Outcome Definition

The v1 thresholds, named suites, and protocol were repository-tagged before hold-out execution. Hard gate and fallback use the same ONNX model and required `CUDAExecutionProvider`; provider fallback or inference error is a technical hard stop. All repetitions of one named condition execute in a fresh CARLA process before the next server. Algorithmic failures are retained.

The campaign contains 2,461 runs in 639 process-isolated sessions. Core has a 66-condition CCRm/CCRB/cut-in speed-gap sweep, 29 mixed vehicle/negative regressions, eight lateral edge props, and two central box/barrel hazards. Six weak four-point synthetic-fault conditions are analysed separately. Local robustness adds 20 speed (± 2 km/h), gap (± 1 m), cut-in-time (± 0.1 s), and prop-offset (± 0.15 m) perturbations. Ten detector-disabled conditions test complete loss of camera detections. The frozen adverse hold-out contains three new central prop types, four vehicle appearances, two new edge props, and five persistent ghost geometries with 6–10 injected points.

Core and hold-out conditions are repeated five times; development and perturbation use two and three repeats. The fixed seed is recorded as 2026. Every reported named cell is consistently all-PASS or all-FAIL over its repeats; thus the named condition—not the run—is the primary descriptive unit. The large run total documents execution coverage, restart

behavior, and repeatability but is not presented as 2,461 independent traffic encounters.

Labels are assigned in frozen YAML before execution. A positive prediction means at least one tick with `aeb_override=true`. Collision means any tick with `collision_count>0`. System PASS requires brake/no-brake and collision to equal their expectations; positive non-collision cases also require minimum bumper gap ≥ 0.5 m, while selected scenarios enforce frozen lane/target assertions. Hence a brake TP may still collide and fail.

We report named-condition TP/FP/TN/FN, PASS, collision, and paired status. Wilson intervals use named-condition counts only and remain finite-grid descriptors, not road-population inference. Suite class ratios are designed. Severity is derived without rerunning CARLA: pre-impact speed is the last 0.05-s tick before first collision; false-brake duration counts override ticks; peak deceleration excludes initialization, sub-1-km/h, and collision ticks. This discrete pre-impact proxy can differ from contact-point speed within one simulation tick and is reported as such. Paired PASS matrices use identical condition identifiers. No null-hypothesis significance test is used because the grids are deliberately constructed rather than randomly sampled from a traffic population.

V. Results

A. Scenario-level core and hold-out trade-off

Table II and Fig. 1 replace run-level uncertainty with named-condition analysis. On core, fallback passes 101/105, versus 99 for hard gate and 93 for radar-only. It restores two non-vehicle conditions missed by hard gating and rejects eight edge-prop conditions that trigger radar-only. The eight radar-only false-positive conditions are four prop types at two lateral offsets; hard gate and fallback reject all. Conversely, radar and fallback brake for both central box/barrel conditions, while the car-only hard gate produces two FN/collisions. All policies also share one FN at a late lateral-corridor cut-out boundary. The three shared collision conditions are two high-speed braking-lead cases and one shortest high-speed cut-in. Hard gate adds the two central-prop collision/FN conditions. Across five consistency repeats this corresponds to 15/25/15 collisions, but those run totals are not independent sample sizes.

Six additional weak four-point synthetic-fault conditions are excluded from primary core confusion: radar-only falsely brakes in all six, while both camera policies block all. Including them would lower radar precision but leave the two camera-policy estimates unchanged; exclusion keeps the primary table focused on physical actors/props and ordinary negative scenarios. In paired core system status, fallback versus hard gate has 99 both-PASS, two fallback-only PASS, zero hard-only PASS, and four both-FAIL conditions. The scenario-level Wilson bounds are visibly wider than the former run-level bounds; for example hard-gate core precision has lower bound 0.955 rather than 0.991. This is the intended statistical correction, not evidence that the point estimate changed.

Hold-out reverses the apparent dominance. It contains three unseen central props, four vehicle appearances, two edge

props, and five persistent 6–10-point synthetic ghosts. Hard gate alone passes four high-support ghost conditions; fallback alone passes none; both pass seven and both fail three. Fallback blocks only the 0.75-m-offset ghost. The ghost returns intentionally reproduce the fallback’s rule-level attributes, so this is a mechanism falsification test, not an estimate of real radar multipath frequency. The label and injection geometry are fixed in YAML: a closing virtual cluster persists near the predicted center path while point count and offset cross the frozen fallback boundaries. Calling this set an “adverse mechanism hold-out” avoids implying an independently sampled deployment distribution.

Vehicle-appearance conditions do not separate hard gate and fallback; the camera label space still covers them. Physical props and ghosts do. This mechanism-level decomposition is more informative than aggregate F1: fallback adds brake recall for unlabelled tracks, but a track-quality rule has no independent evidence of physical existence.

B. Mechanism decomposition

The three physical prop failures occur at different pipeline boundaries. The shopping cart produces at most two path candidates but no confirmed cluster; therefore radar risk never requests a brake and no late permission policy can recover it. This is a radar formation/representation failure, not evidence that the camera veto alone caused impact. The bench does form a confirmed track. Radar-only responds first, fallback waits for simultaneous point, path, TTC, and margin qualification, and hard gating receives no usable car confirmation. The traffic-warning prop is different again: all policies brake at the same gap, and camera-policy logs show confirmation rather than fallback. Despite early permission, configured vehicle/prop contact dynamics still produce collision. The three categories are consequently missing track, late emergency qualification, and collision despite permission.

Ghosts isolate the opposite path through the system. A six-point center injection forms one confirmed track by 0.15 s with confidence 1.0, TTC 0.89 s, and margin -15.21 m; radar-only and fallback immediately request emergency brake, whereas hard gate has no car confirmation. Eight- and ten-point center/0.50-m ghosts behave similarly. Moving the injection to 0.75 m leaves radar-only unchanged but crosses the fallback’s frozen 0.65-m path threshold. Thus the adverse outcome follows directly from the stated rule rather than an unobserved YOLO timing anomaly.

The regression FN is also kept separate. In five repeats of a late cut-out, the radar target leaves/re-enters the predicted lateral corridor around the decision boundary and no policy brakes. Pooling that condition, high-speed stopping-limit collisions, unknown props, and ghosts into one failure rate would hide which stage can plausibly be improved.

C. Ablation, sensitivity, and controlled degradation

A 12-condition development ablation, repeated twice, gives full fallback 12/12. Hard gate and no-latch each fail two

TABLE II
Primary named-condition results; intervals are 95% Wilson descriptors over the constructed condition grid.

Scope	Policy	N	TP	FP	TN	FN	Precision [CI]	Recall [CI]	PASS	Coll.
Core	Radar	105	84	8	12	1	.913 [.838,.955]	.988 [.936,.998]	93	3
	Hard	105	82	0	20	3	1.000 [.955,1]	.965 [.901,.988]	99	5
	Fallback	105	84	0	20	1	1.000 [.956,1]	.988 [.936,.998]	101	3
Hold-out	Radar	14	6	5	2	1	.545 [.280,.787]	.857 [.487,.974]	6	3
	Hard	14	5	0	7	2	1.000 [.566,1]	.714 [.359,.918]	11	3
	Fallback	14	6	4	3	1	.600 [.313,.832]	.857 [.487,.974]	7	3

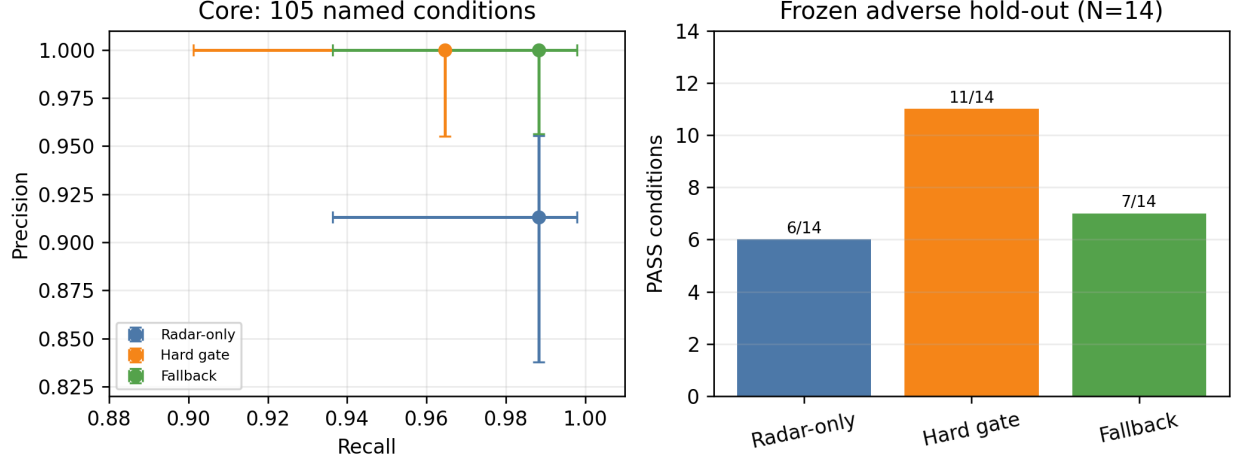


Fig. 1. Named-condition evidence. Left: core precision-recall with descriptive Wilson intervals. Right: system PASS on the 14-condition frozen adverse hold-out. Neither grid represents road prevalence.

TABLE III
Extended named-condition outcomes.

Suite	Policy	Pass/N	TP	FP	TN	FN	Coll.
Perturb.	Radar	19/20	20	0	0	0	1
	Hard	15/20	15	0	0	5	5
	Fallback	18/20	20	0	0	0	1
Camera off	Hard	4/10	0	0	4	6	6
	Fallback	8/10	6	0	4	0	2

central box/barrel conditions. Removing the central-path constraint creates three edge-prop false-brake conditions; removing minimum point count creates two weak-ghost false-brake conditions. Removing stability/confidence or replacing TTC-AND-margin by OR changes none of this focused grid, so independent necessity is not claimed. In six one-factor conditions, lowering point count creates one false brake; increasing stability or tightening TTC each misses one central prop. Other local variants do not change outcomes.

In perturbation, fallback’s 19-m box collides; its 21-m box avoids collision but stops at 0.353 m, below the frozen 0.5-m criterion. This corrects the ambiguity that would result from equating all system FAILs with collisions. With camera disabled, fallback brakes on all six positive conditions but collides in two; brake recall again does not imply safe stopping. Hard gating’s 4/10 camera-off PASS conditions are all true

negatives, so its apparent system rate should not be interpreted as hazard capability.

These grids are one-factor local probes, not global threshold optimization. Specifically, unchanged outcomes for confidence, margin, path, or permissive TTC variants mean only that the selected six conditions did not cross those boundaries. They do not prove the rule is unnecessary. Conversely, the point, path, and latch ablations deliberately contain mechanisms expected to expose them; effect magnitude is therefore suite-dependent.

D. Failure severity and computation

Table IV separates three physical mechanisms. The cart yields no confirmed radar cluster, so all policies impact without braking at 49.96 km/h. The bench yields a track: radar-only brakes at 12.62 m and reaches 32.57 km/h at the last pre-impact tick; fallback waits until 4.29 m and reaches 54.93 km/h; hard gate never brakes and reaches 59.95 km/h. For the warning prop, all policies brake at 21.87 m—including apparent camera confirmation—and reach 7.82 km/h, so a binary collision count hides substantial mitigation.

Every ghost FP is a full simulated stop below 1 km/h, not a one-tick pulse; the last column reports brake duration/peak deceleration for those rows. This makes the binary precision loss operationally meaningful inside simulation, although no following vehicle or occupant model converts it to crash

TABLE IV
Frozen hold-out failure severity (median over five consistent runs).

Mechanism	Policy	Onset / gap	Severity outcome
Cart Bench	All	no brake	49.96 km/h pre-impact
	Radar	12.62 m	32.57 km/h pre-impact
	Hard	no brake	59.95 km/h pre-impact
	Fallback	4.29 m	54.93 km/h pre-impact
Warning	All	21.87 m	7.82 km/h pre-impact
Ghost FP	Radar (25 runs)	78.4 km/h	3.60 s / 9.02 m/s ²
	Fallback (20)	78.4 km/h	3.60 s / 9.02 m/s ²

risk. Conversely, core system-limit collisions occur at approximately 20.43, 37.34, and 77.81 km/h pre-impact under all policies, locating them downstream of the gate. The bench result adds another nuance: radar-only still fails binary system PASS, but its 27.4-km/h reduction relative to hard gate represents substantial impact mitigation that collision count alone would erase. Fallback brakes later and reduces only 5.0 km/h relative to hard gate.

Across 474 required-CUDA sessions, 74,928 inferences complete with zero errors. Session-median p50/p95 are 9.95/11.53 ms; weighted mean is 11.56 ms. Each fresh process has one >150-ms cold start; maximum is 276.10 ms. These are ONNX host processing times under synchronous simulation, not end-to-end ECU deadlines. They exclude radar preprocessing, callback transport, control application, display, and vehicle actuation. The one-per-process cold event is retained rather than discarded as an outlier, because process isolation is part of the protocol.

E. Traceability and audit

Each child summary records scenario/run identifiers, expected and actual brake, collision, minimum gap, first brake time/speed/gap, target association, fallback reason, and failure text. Tick CSV adds point, cluster, confidence, TTC, margin, command, provider, and collision state. Session manifests record commit, seed, CARLA client/server version, model/config SHA-256, configured/active execution provider, inference counts, errors, and latency quantiles. Checkpoint keys are (scenario id, run index), so a technical retry replaces an incomplete run rather than duplicating evidence. The checked derivation script regenerates named confusion, paired outcomes, collision severity, and false-brake severity from the frozen raw archive without changing a run.

VI. Discussion and Validity Boundaries

A. Design implications

The policies encode different error costs. Hard gating is conservative when a car-only detector covers the hazard, but structurally rejects unlabelled objects. Fallback restores selected radar recall while accepting stable, central, high-support ghosts. Track age, point count, confidence, path offset, TTC, and margin are attributes of one radar estimate; none independently proves physical existence. Threshold tuning therefore moves the operating point and can delay genuine braking, as the bench, sensitivity, and camera-off results show.

New evidence—free space, broader semantics, occupancy, calibrated existence probability, or richer radar cues—is needed to break that symmetry.

Policy choice should therefore follow explicit error cost. Where unnecessary high-deceleration braking is dominant, hard gating is favored on this grid; where camera availability or non-vehicle recall dominates, fallback provides partial recovery but requires a stronger existence model. Radar-only remains a useful upper envelope for what the shared radar risk stage can request, not a road-ready policy conclusion. Failure severity should accompany confusion: the warning case shows collision mitigation despite system FAIL, whereas ghost cases show that a false positive can be a complete emergency stop.

B. Validity boundaries

The study’s strength is controlled substitution and adverse evidence, not broad algorithmic novelty. It does not compare learned or probabilistic fusion, confidence-matched soft gates, or multi-class/free-space perception. External validity is limited by one map/ego, one fixed seed, favorable imaging, an in-domain session-separated dataset, point-level CARLA radar, and deterministic repeats. Synthetic ghosts are deliberately constructed, not measured multipath distributions. The adverse hold-out has only 14 designed conditions; its wide intervals make that limitation visible. No weather/friction sweep, hardware-in-the-loop, actuator/communication delay, real radar/camera data, or vehicle test is included. False-brake duration/deceleration and last-tick pre-impact speed improve binary reporting but are not occupant-risk or jerk-cost models. The scenarios do not certify ISO or Euro NCAP compliance.

All child runs store commit, seed, model/config SHA-256, CARLA versions, active provider, and tick-level gate reasons. Raw CSV/JSON, snapshots, manifests, the derivation script, and archive SHA-256 reproduce both run- and scenario-level results. This supports auditability, not road-population inference.

A stronger next protocol should pre-register a stochastic fault surface rather than one threshold-directed ghost: point support 4–12, lateral offset 0–1 m, persistence, range-rate noise, and intermittent returns across multiple seeds. It should add at least one map, weather/illumination change, friction levels, and a second ego platform. Competing policies should include a confidence-matched soft gate, multi-class/free-space visual evidence, and a probabilistic existence tracker. Evaluation should retain named-condition pairing while adding impact speed, brake duration, jerk-aware false-brake cost, and time-to-last-safe-brake objectives. Recorded or hardware-in-the-loop radar would then test whether the mechanism map transfers beyond CARLA abstractions.

VII. Conclusion

On a constructed core grid, emergency fallback combines hard-gate precision and radar-only recall. On a frozen adverse hold-out, hard gating passes more conditions because fallback accepts convincing synthetic ghosts, while all policies

retain distinct missing/late-track physical failures. Scenario-level statistics and severity reveal information hidden by repeated run totals and binary collision counts. Camera gating is therefore false-positive suppression and radar emergency fallback a conditional trade-off—neither is a general safety solution.

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